Tanks or Teachers?

Military Spending in the Solow model

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Motivation



WORLD • NATO

NATO allies agree to raise defense spending to up to 5% of GDP

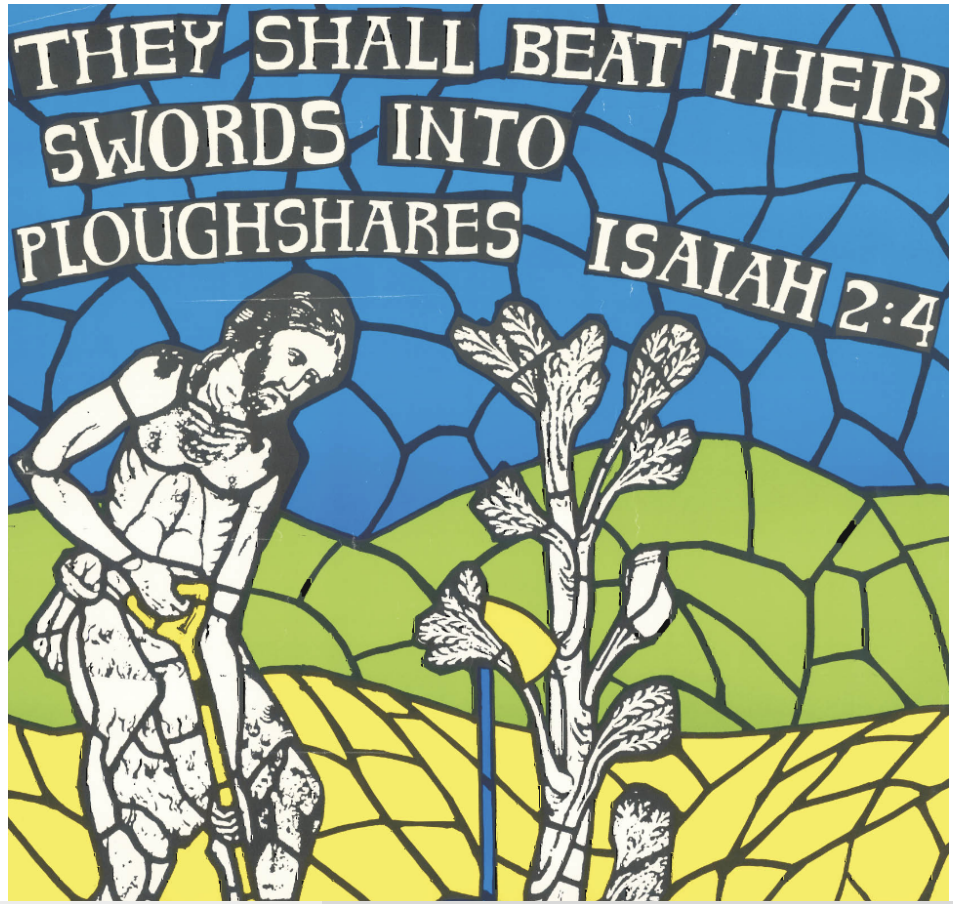
$\Rightarrow$

FRENCH POLITICS • FRANÇOIS BAYROU

Rearmament drive forces France to review budget priorities

The cost of one modern heavy bomber is this: a modern brick school in more than 30 cities. *It is two electric power plants, each serving a town of 60,000 population. It is two fine, fully equipped hospitals. It is some fifty miles of concrete pavement.*
~ Dwight Eisenhower: "Chance for Peace" 1953





Model

normal MRW-type Solow

$$Y = K^{\alpha} H^{\beta} L^{1-\alpha-\beta}$$

State Sector

- fixed tax share tY
  - spent either on education or military
  - $t = e + m$

Role of the Military

higher m => deterrence / protection

- physical capital (destruction)
- human capital (loss of life)

$$z(m) = z_0 e^{-\sigma m}$$

- z_0 = expected war loss (without any military)
  - chance of war * expected destruction
- $e^{-\sigma m}$ = deterrence effect
  - σ = effectiveness of military spending
- $z'(m) < 0$ = decreasing losses
- $z''(m) > 0$ = diminishing returns

Laws of Motion

Now:

$$\dot{k} = \overbrace{s(1-t)y}^{\text{investment}} - \overbrace{(n+d)k}^{\text{depreciation}} - \overbrace{z(m)k}^{\text{destruction}}$$

$$\dot{h} = (t-m)y - (n+d)h - z(m)h$$

Simplification: $D(m) = (n + d + z(m))$

Steady State

Let $\dot{k} = 0$ and $\dot{h} = 0$, then

$$k^* = \frac{s(1-t)}{D(m)} y$$

$$h^* = \frac{(t-m)}{D(m)} y$$

Subst in $y = k^\alpha h^\beta$

$$y^* = \{[s(1-t)]^\alpha [t-m]^\beta D(m)^{-(\alpha+\beta)}\}^{1/1-\alpha-\beta}$$

Solve

Optimum

$$\frac{\overbrace{(\alpha+\beta)\sigma z(m)}^{\text{Marginal Benefit}}}{n+d+z(m)} = \frac{\overbrace{\beta}^{\text{Marginal Cost}}}{(t-m)}$$

LHS = Marginal Benefit of tanks

- Numerator: both capital shares
- Denominator: the D

RHS = Marginal Cost

- Numerator: importance of human capital
- Denominator: share spent on education

$m^* > 0$ if $MB > MC$

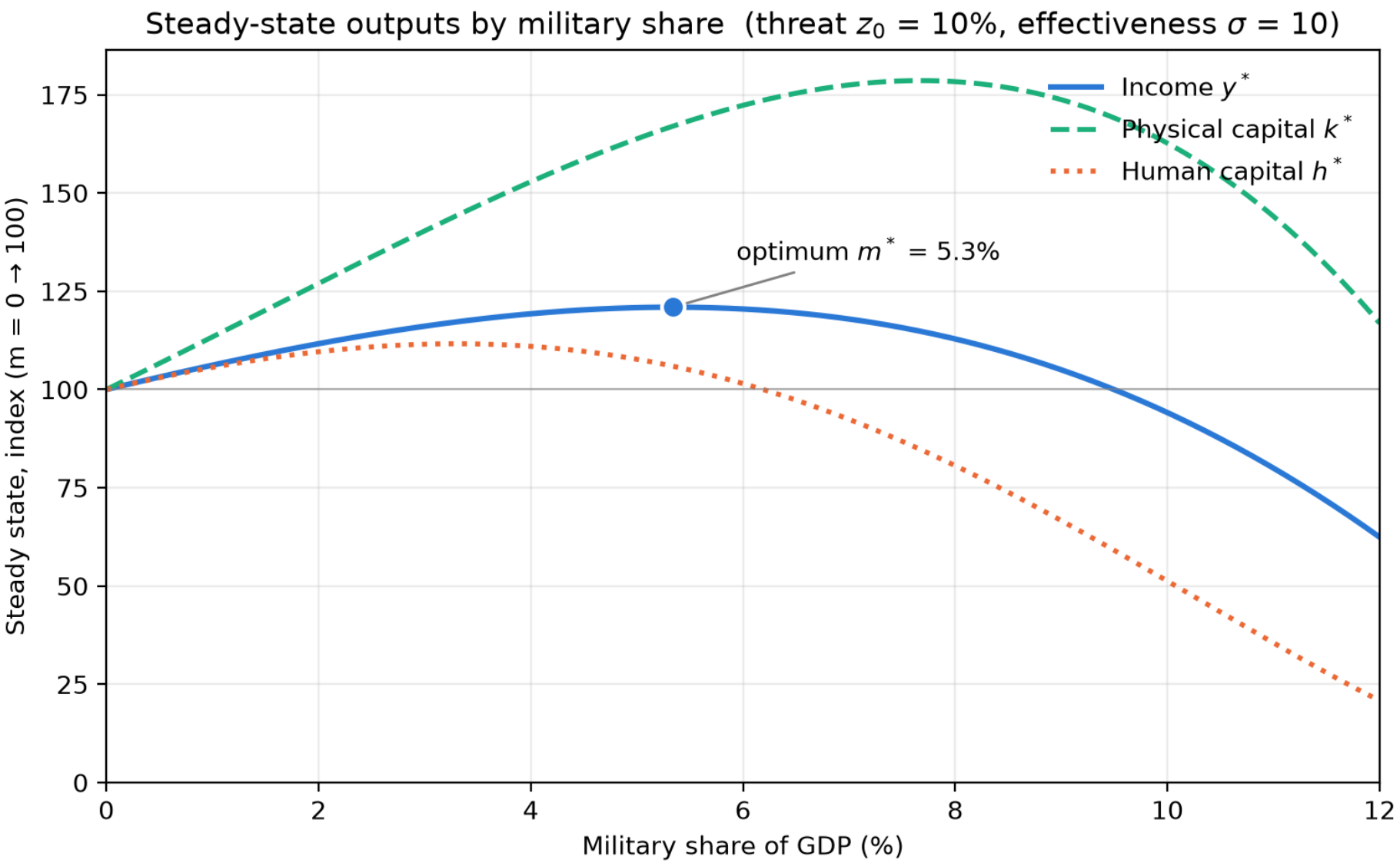
Numerical Example

Illustrative example (because the maths is too complicated)

- $\alpha = \beta = 1/3$
- $n + d = 0.05$
- $t = 0.146 =>$ 14.68% world average tax revenue (World Bank)
- $\sigma = 10$ (1 is never enough)

Threat z0 (%)	Optimal military share m* (% GDP)	Education spending (% GDP)	Income gain vs m = 0 (%)
0.0	0.00	14.68	0.0
5.0	2.96	11.72	5.0
10.0	5.39	9.29	21.3
20.0	7.13	7.55	46.7

Plots


Steady-state outputs by military share (threat z_0 = 10%, effectiveness σ = 10)
Income y^*
Physical capital k^*
Human capital h^*
optimum m^* = 5.3%
Steady state, index (m = 0 → 100)
Military share of GDP (%)
0
25
50
75
100
125
150
175
0
2
4
6
8
10
12

Policy

Why do countries spend different amounts on military?

- different perceived threat level z_0
- higher / lower effectiveness σ (e.g nuclear bomb)

$=$ different political attitudes etc

Note: Assumptions, Rationality, ...

> *Every gun that is made, every warship launched, every rocket fired signifies, in the final sense, a theft from those who hunger and are not fed, those who are cold and are not clothed.*
> $\sim$ ***Dwight Eisenhower**: "Chance for Peace" 1953*